

ARKADELPHIA, AR, USA

WMO: 720394

Lat: 34.100N

Lon: 93.066W

Elev: 56

StdP: 100.66

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00428

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.5	-4.2	-12.7	1.3	-2.3	-10.0	1.6	-0.5	8.5	12.9	7.7	12.2	1.5	330	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	37.7	26.0	36.3	25.9	34.8	25.8	27.9	33.8	27.3	33.2	26.8	32.6	2.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.4	22.0	30.0	26.0	21.6	29.8	25.4	20.7	29.5	88.8	33.7	86.4	33.0	84.1	32.8	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
7.4	6.2	5.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-9.8	38.7	2.3	2.3	-11.4	40.3	-12.8	41.7	-14.1	42.9	-15.7	44.6		
			-10.3	28.8	2.3	0.6	-11.9	29.3	-13.3	29.7	-14.6	30.0	-16.2	30.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	5.5	8.2	13.2	17.5	22.0	26.8	27.8	27.6	24.3	17.5	11.2	7.4
	DBStd	8.94	5.23	5.61	5.02	3.86	3.46	2.29	2.42	2.57	3.23	4.56	4.93	4.91
	HDD10.0	438	157	93	26	2	0	0	0	0	3	44	113	
	HDD18.3	1554	398	286	170	59	11	0	0	0	1	71	218	339
	CDD10.0	3165	17	43	126	228	371	504	551	547	430	236	80	33
	CDD18.3	1239	1	2	12	34	124	254	293	288	181	46	4	1
	CDH23.3	13056	1	11	109	334	1083	2779	3277	3139	1759	512	46	6
	CDH26.7	6354	0	0	14	60	384	1429	1743	1677	856	186	6	0
(14) Wind	WSAvg	1.8	2.2	2.4	2.5	2.3	1.8	1.7	1.3	1.2	1.3	1.5	1.7	2.0
(15) Precipitation	PrecAvg	1418	113	104	134	129	156	106	106	71	108	139	116	130
	PrecMax	2228	246	223	303	271	399	316	330	240	289	410	293	274
	PrecMin	878	4	18	41	55	57	25	5	10	30	2	28	15
	PrecStd	279	65	46	69	62	73	71	76	45	62	96	62	60
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.7	24.9	28.1	29.9	33.0	37.9	38.8	41.1	37.6	33.0	27.6	24.0
		MCWB	16.5	18.5	20.0	21.5	24.5	25.3	26.2	26.7	24.6	22.3	21.0	21.1
	2%	DB	19.8	22.5	26.4	28.0	32.0	36.2	37.5	37.9	35.6	31.0	24.1	21.2
		MCWB	15.3	17.9	18.7	20.2	23.9	25.9	26.3	26.4	24.4	22.1	18.3	18.7
	5%	DB	17.4	20.4	23.9	27.2	30.6	34.5	36.1	36.2	33.5	28.7	22.3	18.8
		MCWB	13.6	16.5	17.5	19.8	23.3	25.6	26.4	26.0	24.3	21.1	18.2	16.1
	10%	DB	14.6	17.9	22.1	25.5	28.9	33.2	34.1	34.1	31.9	26.3	19.9	16.2
		MCWB	10.8	14.8	16.7	19.2	22.7	25.3	26.2	25.9	24.0	20.4	16.8	13.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	20.3	21.5	23.9	26.4	28.0	28.5	28.6	27.0	25.6	22.6	21.7
		MCDB	21.4	22.8	26.5	27.4	31.3	33.6	34.5	35.1	32.5	30.4	25.4	22.2
	2%	WB	17.0	19.0	20.4	22.4	25.3	27.3	27.8	27.9	26.2	24.4	20.5	19.2
		MCDB	18.7	21.3	24.8	26.2	30.0	32.9	33.7	34.1	31.7	28.8	23.0	20.8
	5%	WB	14.4	17.4	19.0	21.4	24.5	26.7	27.3	27.4	25.5	22.9	19.0	16.8
		MCDB	16.1	19.3	22.5	25.0	28.7	32.5	33.2	33.4	31.1	26.2	21.3	17.9
	10%	WB	11.6	15.4	17.7	20.4	23.8	26.0	26.8	26.8	24.9	21.4	17.2	14.0
		MCDB	13.9	17.6	20.8	23.8	27.5	31.7	32.7	32.6	30.1	24.3	19.3	15.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	11.4	12.6	12.6	11.4	12.0	11.7	12.0	13.0	14.1	13.0	11.0
		MCDBR	16.1	14.4	15.0	14.8	13.3	13.8	14.5	15.2	15.7	17.1	15.5	14.5
		MCWBR	11.3	10.0	8.5	7.7	6.1	4.9	5.0	5.2	6.0	8.5	10.1	11.6
	5% WB	MCDBR	12.9	11.6	12.3	11.8	11.4	12.0	12.4	13.0	12.6	13.0	12.3	12.7
		MCWBR	11.0	9.0	8.0	7.3	5.8	5.0	5.0	5.3	5.4	7.5	9.2	11.2
(40) Clear-Sky Solar Irradiance	taub		0.329	0.334	0.366	0.399	0.434	0.466	0.488	0.484	0.414	0.364	0.341	0.319
	taud		2.491	2.476	2.409	2.317	2.281	2.219	2.187	2.188	2.361	2.477	2.494	2.529
	Ebn at Noon		880	915	908	888	857	825	804	802	847	870	859	871
	Edh at Noon		86	96	111	128	134	143	146	144	116	95	85	78
(44) All-Sky Solar Radiation	RadAvg		2.55	2.95	4.15	5.19	5.67	6.33	6.20	5.74	4.92	3.86	2.76	2.23
	RadStd		0.23	0.45	0.35	0.33	0.48	0.54	0.39	0.46	0.41	0.54	0.32	0.26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page